RO Skid - HIL Test Report

Validation Results

EQSP32 Industrial IoT Controller

Document	Detail
Title	Reverse Osmosis Skid - Hardware-in-the-Loop Test Report
Date	2026-06-15 10:47:36
Device	EQSP32 (ESP32-S3) on COM35, firmware EQSP32_RO_Skid_Controller (HIL build)
Specification	RO Skid - Control & Operation Specification v0.3
Test plan	RO Skid - HIL Test Plan v1.0
Result	31 / 31 checks passed (ALL PASS)

ALL 31 CHECKS PASSED

Method

All sensors were injected and all outputs read back over the EQSP32 Hardware-in-the-Loop serial protocol on COM35. Controller state, status-LED indication and actuator outputs were verified from the firmware's >>STATE telemetry and from output-cache queries. Long process timers were time-compressed for the HIL build (flush interval 8 s, flush 2 s, TDS trip 3 s, settle / min-on / min-off / dry-run grace 1.5 s); the control logic and all setpoints are identical to production. See the HIL Test Plan for the full methodology and scaling.

Results

#	Test	Expected	Actual	Result
1	A1: Run/Stop=STOP with healthy sensors -> safe STOP, LED off	st=STOP pump=0 sv=0 fv=0 led=OFF	st=STOP pump=0 sv=0 fv=0 led=OFF	PASS
2	A2: Run/Stop=RUN -> auto-start through STARTING to RUNNING (no operator start)	st=RUNNING pump=1 sv=1 fv=0 led=GREEN	st=RUNNING pump=1 sv=1 fv=0 led=GREEN	PASS
3	A3: Output cache cross-check (pump/supply/flush POUT duty)	EQ.15=1000 EQ.16=1000 EQ.14=0	EQ.15=1000 EQ.16=1000 EQ.14=0	PASS
4	B1: Clean tank full (>=400) -> stop, IDLE_FULL, blue solid	st=IDLE_FULL pump=0 led=BLUE/SOLID	st=IDLE_FULL pump=0 led=BLUE blink=SOLID	PASS
5	B2: Clean 385 (between 380-400) -> hysteresis holds, stays stopped	st=IDLE_FULL pump=0	st=IDLE_FULL pump=0 cln=385	PASS
6	B3: Clean falls to 375 (<=380) -> auto restart to RUNNING	st=RUNNING pump=1	st=RUNNING pump=1 cln=375	PASS

7	B4: Supply tank low (<50) -> pause, PAUSE_SUPPLY, blue blink	st=PAUSE_SUPPLY pump=0 led=BLUE/BLINK	st=PAUSE_SUPPLY pump=0 led=BLUE blink=BLINK	PASS
8	B5: Supply 55 (between 50-60) -> hysteresis holds, stays paused	st=PAUSE_SUPPLY pump=0	st=PAUSE_SUPPLY pump=0 sup=55	PASS
9	B6: Supply rises to 65 (>=60) -> auto resume to RUNNING	st=RUNNING pump=1	st=RUNNING pump=1 sup=65	PASS
10	C1: Feed pressure >200 with normal flow -> early-fouling WARNING (yellow slow)	warn=HP led=YELLOW/SLOW	warn=HP led=YELLOW blink=SLOW pin=210	PASS
11	C2: Feed pressure >200 with low permeate -> CLEANING REQUIRED (yellow fast)	warn=CLEAN led=YELLOW/FAST	warn=CLEAN led=YELLOW blink=FAST pin=210 perm=1.8	PASS
12	C3: Low feed flow (<8 L/min) -> flow WARNING (yellow medium)	warn=FLOW led=YELLOW/MED	warn=FLOW led=YELLOW blink=MED feed=4.6	PASS
13	C4: Recovery out of band (>30%) -> recovery WARNING (yellow medium)	warn=REC led=YELLOW/MED	warn=REC led=YELLOW blink=MED rec=35	PASS
14	C5: Rising permeate TDS (>160, <200) -> TDS WARNING (yellow fast)	warn=TDS led=YELLOW/FAST	warn=TDS led=YELLOW blink=FAST tdso=180	PASS
15	D1: Motor over-temperature (>=80C) -> FAULT, red, pump off	fault=TEMP fcode=1 pump=0 led=RED	st=FAULT fault=TEMP fcode=1 pump=0 led=RED	PASS
16	D1c: Run/Stop=STOP clears the over-temperature fault	fault=NONE st=STOP	st=STOP fault=NONE	PASS
17	D2: Dry-run / no feed flow (>grace) -> FAULT, code 2	fault=DRYRUN fcode=2 pump=0	st=FAULT fault=DRYRUN fcode=2 pump=0	PASS
18	D2c: Run/Stop=STOP clears the dry-run fault	fault=NONE st=STOP	st=STOP fault=NONE	PASS
19	D3: Permeate TDS >200 sustained (>=10 min, test 3s) -> FAULT, code 3	fault=TDS fcode=3 pump=0	st=FAULT fault=TDS fcode=3 pump=0 tdso=210	PASS
20	D3c: Run/Stop=STOP clears the TDS fault	fault=NONE st=STOP	st=STOP fault=NONE	PASS
21	D4: Over-pressure (>250 PSI) -> FAULT, code 4	fault=PRESSURE fcode=4 pump=0	st=FAULT fault=PRESSURE fcode=4 pump=0 pin=260	PASS
22	D4c: Run/Stop=STOP clears the over-pressure fault	fault=NONE st=STOP	st=STOP fault=NONE	PASS
23	D5: Analog sensor broken wire (<3.5 mA) -> sensor FAULT, code 5	fault=SENSOR fcode=5 pump=0	st=FAULT fault=SENSOR fcode=5 pump=0	PASS
24	D5c: Run/Stop=STOP clears the sensor fault	fault=NONE st=STOP	st=STOP fault=NONE	PASS
25	E1: E-stop while RUNNING -> immediate safe STOP (all outputs off, LED off)	st=STOP pump=0 sv=0 fv=0 led=OFF	st=STOP pump=0 sv=0 fv=0 led=OFF	PASS
26	E2: E-stop also clears a latched fault (STOP, fault=NONE)	st=STOP fault=NONE led=OFF	st=STOP fault=NONE led=OFF	PASS
27	F1: Minimum pump ON-time honoured (full tank does not stop pump before min-on)	pump=1 @0.8s then pump=0 @2.3s	pump@0.8s=1 pump@2.3s=0	PASS
28	F2: Minimum pump OFF-time honoured (restart waits min-off after a stop)	pump=0 @0.8s then pump=1 @2.4s	pump@0.8s=0 pump@2.4s=1	PASS
29	G1: Periodic flush after run interval -> FLUSHING (pump on, flush valve open, yellow solid)	st=FLUSHING pump=1 fv=1 led=YELLOW/SOLID	st=FLUSHING pump=1 fv=1 sv=1 led=YELLOW blink=SOLID	PASS
30	G2: During flush the brine-bypass (flush) valve output is energised	EQ.15=1000 EQ.14=1000	EQ.15=1000 EQ.14=1000	PASS
31	G3: Flush completes -> returns to RUNNING, flush valve closes	st=RUNNING fv=0 led=GREEN	st=RUNNING fv=0 led=GREEN	PASS

Coverage

- A - Stopped / auto-run: power-on auto-start with no operator action, safe stop, and output-cache cross-check.
- B - Tank logic: clean-tank-full stop and supply-low pause, both with hysteresis (no short-cycling at the thresholds).
- C - Production warnings: high feed pressure, membrane-cleaning-required, low flow, recovery out of band and rising TDS, each with the correct yellow flash rate.
- D - Latched faults: over-temperature, dry-run, permeate TDS > 200, over-pressure and broken-wire sensor, each with its red blink code and Run/Stop reset.
- E - Emergency stop: immediate safe state and fault clear on STOP.
- F - Motor anti short-cycle: minimum ON-time and minimum OFF-time.
- G - Periodic membrane forward-flush and return to production.

Conclusion

The firmware EQSP32_RO_Skid_Controller meets the behaviour defined in the RO Skid Control & Operation Specification v0.3 across all 31 validated checks. The build is accepted for the on-bench (HIL) stage; the next step is commissioning against the physical skid with the production timer values.